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$$PV = nRT$$

or

$$P = \frac{n}{V}RT$$

Since

Active mass = molar concentration = n/V

P = active mass × RT

Based on the above expression the partial pressure of the reactants and products can be expressed as,

$$p_A^x = [A]^x [RT]^x$$

$$p_B^y = [B]^y [RT]^y$$

$$p_C^l = [C]^l [RT]^l$$

$$p_D^m = [D]^m [RT]^m$$

On substitution in Eqn. 2,

$$k_p = \frac{[C]^l [RT]^l [D]^m [RT]^m}{[A]^x [RT]^x [B]^y [RT]^y} \qquad (3)$$

$$k_p = \frac{[C]^l [D]^m [RT]^{l+m}}{[A]^x [B]^y [RT]^{x+y}}$$

$$k_p = \frac{[C]^l [D]^m}{[A]^x [B]^y} [RT]^{(l+m)-(x+y)} \qquad (4)$$

By comparing equation (1) and (4), we get

$$k_p = K_c (RT)^{(\Delta n_g)} \qquad (5)$$

where,

Δn_g is the difference between the sum of number of moles of products and the sum of number of moles of reactants in the gas phase.

The following relations become immediately obvious.

When $\Delta n_g = 0$

$K_p = K_c (RT)^0 = K_c$

Example:

$H_2(g) + I_2(g) \rightleftharpoons 2HI\ (g)$

$N_2\ (g) + O_2\ (g) \rightleftharpoons 2NO(g)$

When Δn_g = +ve

$K_p = K_c (RT)^{+ve}$

$K_p > K_c$

$2NH_3(g) \rightleftharpoons N_2\ (g) + 3H_2\ (g)$

$PCI_5\ (g) \rightleftharpoons PCl_3\ (g) + Cl_2\ (g)$

When Δn_g = -ve

$K_p = K_c (RT)^{-ve}$

$K_p < K_c$

Example:

$2H_2(g) + O_2(g) \rightleftharpoons 2H_2O\ (g)$

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3\ (g)$

Table 8.1 Relation between equilibrium constants for some reversible reactions

Reversible reactions	Equilibrium constant
For the forward reaction $x\,A + yB \rightleftharpoons lC + mD$	K_c